7.1.1 Binomial distribution

Binomial distribution was discovered by James Bernoulli(1654-1705) in the year 1700 and was first published posthumously in 1713 , eight years after his death.

A random experiment whose outcomes are of two types namely success S and failure F, occurring with probabilities p and q respectively, is called a Bernoulli trial.

Some examples of Bernoulli trials are :

(i) Tossing of a coin (Head or tail)

(ii) Throwing of a die (getting even or odd number)

Consider a set of n independent Bernoullian trails (n being finite) in which the probability 'p' of success in any trial is constant , then $q = 1-p$, is the probability of failure. The probability of x successes and consequently $(n-x)$ failures in n independent trials, in a specified order (say) SSFSFFFS….FSF is given in the compound probability theorem by the expression

$$\begin{aligned} P(SSFSFFFS.....FSF) &= P(S)P(S)P(F)P(S)\ldots \\ &\quad P(F)P(S)P(F) \\ &= p.p.q.p \ldots q.p.q \\ &= p.p.p.p \ldots q.q.q.q.q.q \ldots \\ &= \{x \text{ factors}\}\ \{(n-x) \text{ factors}\} \\ &= p^x q^{(n-x)} \end{aligned}$$

x successes in n trials can occur in ${}^nC_x$ ways and the probability for each of these ways is same namely $p^x q^{n-x}$.

The probability distribution of the number of successes, so obtained is called the binomial probability distribution and the binomial expansion is $(q+p)^n$

Definition 7.1

A random variable X is said to follow binomial distribution with parameter n and p, if it assumes only non- negative value and its probability mass function in given by

$$P(X = x) = p(x) = \begin{cases} {}^nC_x p^x q^{n-x}, & x = 0,1,2,\ldots..n; q = 1-p \\ 0 & , otherwise \end{cases}$$

Note

Any random variable which follows binomial distribution is known as binomial variate i.e $X \sim B(n,p)$ is a binomial variate.

The Binomial distribution can be used under the following conditions :

1. The number of trials 'n' finite
2. The trials are independent of each other.
3. The probability of success 'p' is constant for each trial.
4. In every trial there are only two possible outcomes – success or failure.

Derivation of the Mean and Variance of Binomial distribution :

The mean of the binomial distribution

$$\begin{aligned} E(X) &= \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x} \\ &= p \sum_{x=1}^{n} x. \binom{n}{x} \binom{n-1}{x-1} p^{x-1} q^{n-x} \\ &= np\ (q+p)^{n-1} \qquad [\text{since } p+q = 1] \\ &= np \end{aligned}$$

$$E(X) = np$$

$\therefore$ The mean of the binomial distribution is np

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

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